INCH-POUND
MIL-DTL-24643/38G
27 November 2023
SUPERSEDING
MIL-DTL-24643/38F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LSECM

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This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight)

Component 1: 8 shielded pairs

First - Copper conductor, tin-coated, in accordance with ASTM B286, 20 AWG with 7 strands.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Thermoset insulation. Nominal wall thickness of 0.023 inch; wall thickness may vary from nominal as necessary, in order for the completed cable to meet the specified electrical requirements. Colored insulation, one white and one black conductor for each pair.

Fourth - Two conductors (one black and one white) cabled together with a nominal lay of 2.5 inches to form a pair.

Fifth - An optional binder.

Sixth - Braided shield of 34 or 36 AWG, tin-coated copper. Braid angle of 20 to 40 degrees; minimum coverage of 85 percent.

Seventh - Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape, plus an extruded jacket of a suitable transparent material; minimum average thickness of the transparent material 0.003 inch. Standard identification code shall be applied by method 2.

Component 2: 8 groups (seven conductors in each)

First - Copper conductor, coated or uncoated, 18 AWG, class B stranding (7 by 0.0152 inches). Maximum diameter of 0.050 inch.

Second - Thermoset insulation. Nominal wall thickness of 0.008 inch. Colored insulation, five black and two white conductors for each group.

Third - Seven conductors, five black and two white, with the white conductors adjacent in the layer of six, cabled with a lay not greater than 2 inches, to form each group.

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Fourth - Group covering of two polyester tapes, outer tape sealed. An alternate group covering may consist of one polyester tape, plus an extruded jacket of a suitable transparent material; minimum average thickness of the transparent material 0.003 inch. Standard identification code shall be applied by method 2.

Assembly

First - Five pairs cabled to form a core. Three pairs and eight groups of seven cabled over the core, with a lay in accordance with 3.4.4 of MIL-DTL-24643. Pair no. 1 through 5 in core; the sequence of the outer layer – pair no. 6: groups no. 1, 2, and 3; pair no. 7: groups no. 4, 5, and 6; pair no. 8: groups no. 7 and 8. Fillers may be used as necessary to form a firm, well-rounded assembly.

Second - An optional binder.

Third - Cross-linked polyolefin jacket. See table I for jacket thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

TABLE I. Details.

Military part no. M24643/38	Type	Number of conductors in pairs	Number of conductors in groups	Cable jacket thickness min. avg. (inch)	Overall diameter (inches)	
					min.	max.
-01UN	LSECM	16	56	0.075	1.290	1.370

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)

ASTM B286, 20 AWG with 7 strands	11.03
18 AWG, class B	7.47

Voltage withstand (volts, root mean square, min.)

Conductor to conductor	2,000
Conductor to shield (pairs only)	1,000
Component shield to component shield	500

Insulation resistance (megohms/1,000 feet, min.)

Conductor to conductor	500
Conductor to shield (pairs only)	500
Component shield to component shield	100

Conductor continuity	No failure
Shield continuity	No failure

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Group A:	
Visual and dimensional	No failure
Watertightness (see MIL-DTL-24643 for limits of water leakage)	No failure
Capacitance (pairs only)	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 feet)	30
Group B:	
Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Drip (203±1.8 °F [95±1 °C])	No failure
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Shrinkage	No failure
Cable fill (sealant) removability	No failure

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Shield (conformance to material, construction, and coverage)	No failure

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation/group covering	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

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UNIT ORDERING LENGTHS:

Type	Feet (nominal)
LSECM	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-054)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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